

Question number	Answer	Additional guidance	Mark
6(a)	A description that includes three of the following points: • whether it caused side effects (in humans) (1) • whether or not it reduced the size of the plaques (1) • {(an effective) concentration / the half life} of the drug (1) • the (best) method to administer the drug (1) • how long to administer drug (for amyloid to be destroyed) (1)	ACCEPT named side effect e.g. allergic reaction ACCEPT {plaque / amyloid} disappeared better than existing drugs effective ACCEPT dose	(3)

Question number	Answer	Additional guidance	Mark
6(b)(i)	An explanation that includes the following points: • antibodies can <u>only</u> bind to their antigen / this antibody can <u>only</u> bind to β amyloid (1) • because of the (complementary) shape of {antigen / antibody} (1)	ACCEPT FAB region / FAB receptor / receptor	(2)

Question number	Answer	Additional guidance	Mark
6(b)(ii)	An answer that includes the following points: • {ATD / antibodies} will bind to the <u>amyloid</u> (1) • macrophages engulf the amyloid (1) • macrophages {destroy / digest} the amyloid with {enzymes / lysozymes} (1)	ACCEPT opsonisation ACCEPT phagocytes / neutrophils ACCEPT phagocytes / neutrophils protease DO NOT ACCEPT lysosome	(3)

Question number	Answer	Additional guidance	Mark
6(c)(i)	An answer that includes the following points: • because their immune system had already been activated (against amyloid) (1) • and therefore {the amyloid was broken down / plaques could not form} (by phagocytes) (1)	ACCEPT a description e.g. a primary immune response had already taken place, the person had memory cells (to amyloid) antibodies present DO NOT ACCEPT by antibodies / B cells	(2)

Question number	Answer	Additional guidance	Mark
6(c)(ii)	An explanation that includes the following points: - because people with Alzheimer's disease will not have these memory cells (1) - and therefore there will not be any memory cells that can differentiate into the plasma cells that produce the antibody (1)	ACCEPT B memory cells not effective ACCEPT no memory cells that will result in the production of antibody DO NOT ACCEPT no memory cells to produce the antibody	(2)